

Requirement Analysis Document (RAD)

Project Title: ShopSphere Online Shopping Platform

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1. Executive Project Summary

ShopSphere is an e-commerce platform similar to Amazon and Flipkart. It enables customers to register, browse products, search items, manage shopping carts, place orders, make secure payments and track deliveries. The system uses an RDBMS and follows ACID properties to ensure reliable and accurate transaction processing.

2. Core Business Domains & Features

Customer Registration: Unique email, login, profile and address management.

Product Management: Categories, brands, pricing, stock and availability.

Shopping Cart & Orders: Add/remove items, checkout and order lifecycle.

Payment Management: UPI, Cards, Net Banking, COD and payment status.

Shipment Tracking: Packed, shipped, in transit and delivered.

3. Actor Profiles

Customer: Browse, purchase and review products.

Seller: Manage products and inventory.

Delivery Partner: Update shipment status.

Administrator: Manage users, products, orders and security.

4. Reporting Requirements

Best Selling Products Report, Customer Purchase Report, Monthly Sales Report, Inventory Report and Payment Report.

5. Non-Functional Requirements

Secure login, encrypted passwords, unique email IDs, positive price validation, stock validation, concurrency control, audit logs, fast search response, reliable backup and recovery.

Conclusion

This Requirement Analysis Document defines the functional and non-functional requirements of the ShopSphere Online Shopping Platform and serves as the foundation for database design and system development.